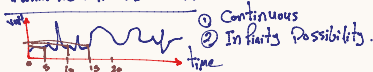
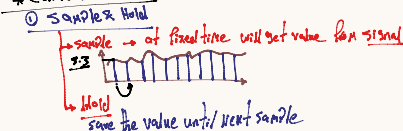


ADC (Analog to Digital Converter)

* what mean Analog signal?

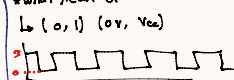


* Convert Process



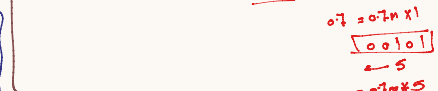
- 1) Note → 1) ADC → change from Analog volt to digital volt
- 2) ant-sensor → transducer from Power to electricity.

* what mean Digital?



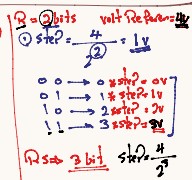
2) Quantization

- 1) Convert the value from analog to digital
- 2) Relation (step) = $\frac{\text{Vref}}{\text{Resolution}}$



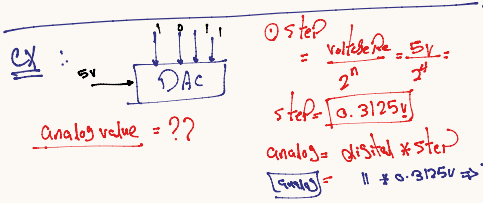
* Quantization → change to digital value (digital volt)

- Resolution → # No of bits
- No of bits 4 →

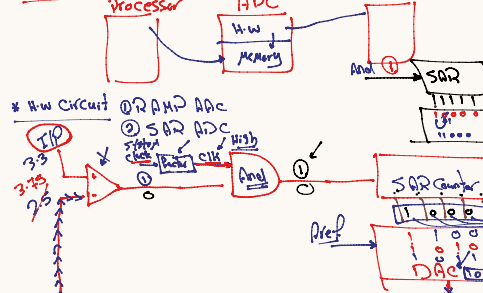


Note: Quantization Error → can't detect the new value

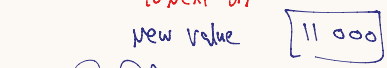
- 1) Max value bit digital = voltage reference - one step



* ADC → Peripheral processor



* SAR Counter Receive 1 from Analog gate



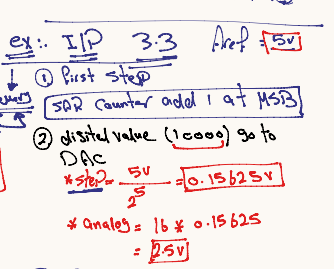
* DAC

- * step = 0.15625
- * analog = 24 * step = 24 * 0.15625 = 3.75

* OP AMP

- * new value 10100 → 3.125

* OP AND → electronic Component



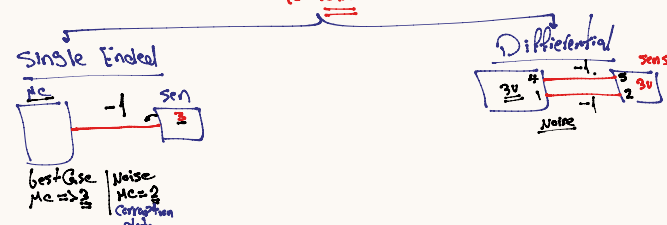
Inp 3.3 > Ref 2.5



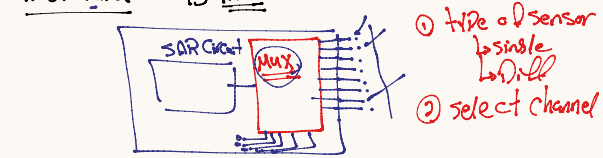
* Sensor

→ Digital sensor → (0, 1) → IR

→ Analog sensor → Convert from sensing to volt



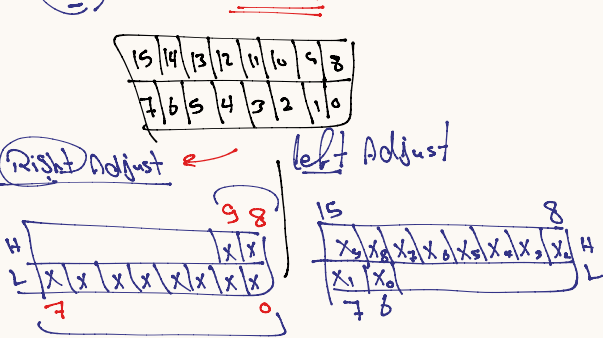
* Channel → 1) ADC



* Value very small → Gain →

* Data size at least 16 bit = 2^16

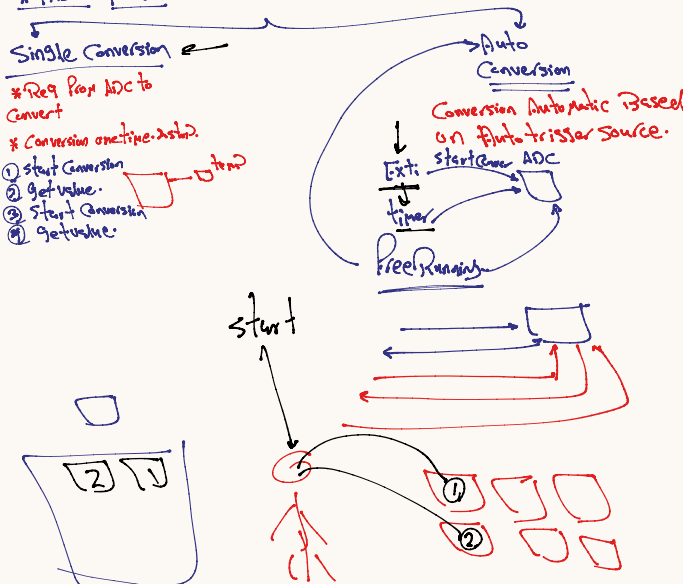
* H.W write the Result



* ADC

- 1) ADC circuit need Isolate Power to work. (Avee) Int kit → the Avea Connect with 5V Direct
- 2) ADC circuit need the voltage Reference to DAC work → Aref

* ADC Mode



1) The one Conversion take the no of bit times.

→ no of compare = no of bits

2) You Can change → Reference voltage / Input / clk



Memory

- | | |
|----------|-----------|
| ① AD_MUX | ② ADC_SRA |
| ③ ADCH | ④ ADCL |
| ⑤ SPIOR. | |

*AD_MUX

- select voltage Reference
- select Adjust Result
- select channel / type sensor